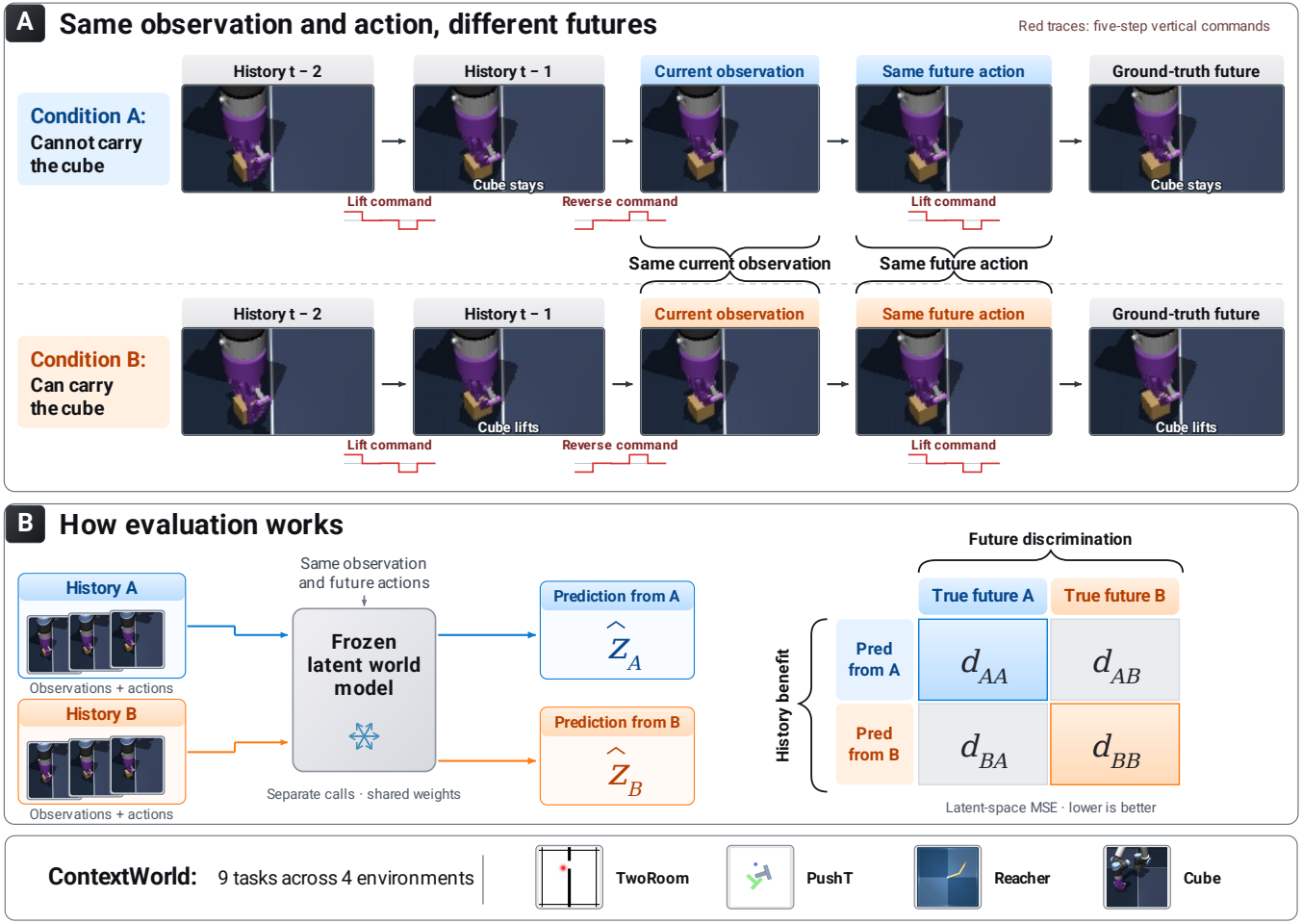

A
Same observation and action, different futures
Red traces: five-step vertical commands
Condition A:
Cannot carry the cube
History t − 2
History t − 1
Current observation
Same future action
Ground-truth future
Cube stays
Lift command
Reverse command
Lift command
Cube stays
Same current observation
Same future action
Condition B:
Can carry the cube
History t − 2
History t − 1
Current observation
Same future action
Ground-truth future
Cube lifts
Lift command
Reverse command
Lift command
Cube lifts
B
How evaluation works
History A
Observations + actions
History B
Observations + actions
Same observation and future actions
Frozen latent world model
Separate calls · shared weights
Prediction from A
$\hat{z}_A$
Prediction from B
$\hat{z}_B$
Future discrimination
True future A
True future B
History benefit
Pred from A
Pred from B
d_{AA}
d_{AB}
d_{BA}
d_{BB}
Latent-space MSE · lower is better
ContextWorld:
9 tasks across 4 environments
TwoRoom
PushT
Reacher
Cube